

SODIUM

HYPONATREMIA

CRITICAL LOW: not stated; LOW <135

HYPERNATREMIA

CRITICAL HIGH not stated; >145

RANGE: - 135-145 mEq/L

What does it do in the body? What body system/organ(s) favors it? What is the expected function for that body system(s)? * Note: Understanding function helps understand the problems when it is low/high *		
- Major extracellular ion -> controls water balance / osmolality		
- Maintains blood volume + BP; supports nerve impulses		
- Kidneys, ADH, and aldosterone regulate the level		
IMPLICATIONS & NURSING MANAGEMENT		
What causes it to DECREASE?		What causes it to INCREASE?
- Na loss: vomiting, NG suction, sweating		- Water loss: fever, sweating, vomiting, diarrhea
- Diuretics / renal loss; low intake or NPO		- Diabetes insipidus -> free-water loss
- CHF or hypotonic IV fluid -> dilution		- Excess Na: oral/IV, tube feed, hypertonic fluid
- SIADH -> excess ADH -> water retention		- Corticosteroids / aldosterone excess
Signs & Symptoms when it's LOW? Critically LOW?		Signs & Symptoms when it's HIGH? Critically HIGH?
- Confusion, lethargy, poor concentration		- Thirst; dry mouth/skin; fever or flushed skin
- Seizures or stupor = severe neurologic danger		- Agitation or confusion
- Weakness; shallow respirations can be late		- Edema / fluid retention when Na gain is the cause
- Abdominal cramps/low BS; low BP and urine output		- Low urine output; dehydration findings with water loss
How do we fix it?		How do we fix it?
- Trend Na; neuro/VS/RR/I&O; seizure precautions		- Trend Na; neuro/VS/I&O/daily wt; fall precautions
- SIADH/dilution: fluid restrict +/- diuretic as ordered		- Ordered isotonic or hypotonic IV fluid
- Hypertonic IV as ordered; watch fluid overload		- Correct slowly -> rapid drop risks cerebral edema
- Treat cause; moderate salty foods only if appropriate		- Restrict Na; treat cause (example: DI)
ENCOURAGE	DIETARY SOURCES (HIGHEST CONTENT PER SERVING)	AVOID
Soy sauce, canned soups, deli cold cuts, frozen "tv" dinners, fried foods, pickles, bacon,		
LOW: ENCOURAGE sources above only if appropriate		HIGH: LIMIT / AVOID sources above
(arranged from higher content to lower but all rich in sodium.)		

HYPOKALEMIA

CRITICAL LOW: not stated; LOW <3.5; <2.5 urgent

POTASSIUM

RANGE: - 3.5-5.0 mmol/L

HYPERKALEMIA

CRITICAL HIGH not stated; >5.0

What does it do in the body? What body system/organ(s) favors it? What is the expected function for that body system(s)?
* Note: Understanding function helps understand the problems when it is low/high *

- Major intracellular ion -> nerve impulse transmission
- Controls skeletal + cardiac muscle contraction
- Heart rhythm is priority; kidneys regulate the level

IMPLICATIONS & NURSING MANAGEMENT

What causes it to DECREASE?

- GI loss: vomiting, diarrhea, NG drainage
- Loop/thiazide diuretics; poor intake/dehydration
- Insulin or albuterol -> K moves into cells
- Hyperaldosteronism / corticosteroids

Signs & Symptoms when it's LOW? Critically LOW?

- Weak/irregular pulse; low BP or orthostasis
- Weakness/cramps; low DTR; paralysis if severe
- Shallow RR; diminished breath sounds; low bowel sounds
- ECG: ST depression, inverted T, prominent U wave

How do we fix it?

- Telemetry; monitor VS/RR/GI/DTR; check K + Mg
- Check renal function + urine output before replacement
- Oral K with food when possible
- IV K: dilute/pump/slow; monitor site - NEVER IV push

What causes it to INCREASE?

- Renal impairment -> decreased K excretion
- ACE inhibitor, spironolactone, or NSAID
- Addison disease -> too little aldosterone
- Burns/trauma (cell lysis) or excess IV K

Signs & Symptoms when it's HIGH? Critically HIGH?

- Irregular/brady pulse; low BP; VF if severe
- ECG: peaked T waves and ST elevation
- Weakness/cramps; paresthesia; low DTR
- Respiratory failure; hyperactive BS/diarrhea; confusion

How do we fix it?

- Continuous ECG; stop K; question K-retaining meds
- IV calcium gluconate -> protects the heart
- Insulin+dextrose / albuterol / bicarb -> shift K in
- Diuretic, K binder, or dialysis -> removes K

ENCOURAGE

DIETARY SOURCES (HIGHEST CONTENT PER SERVING)

AVOID

Baked potato, spinach, avocado, white beans, sweet potato, tomato sauce, banana, orange juice, lentils

LOW: ENCOURAGE potassium foods above

HIGH: LIMIT foods; NO salt substitutes

(arranged from higher content to lower but all rich in potassium.)

HYPOCALCEMIA

CRITICAL LOW: not stated; LOW <9.0

CALCIUM

RANGE: - 9-11 mg/dL*

HYPERCALCEMIA

CRITICAL HIGH not stated; >11

What does it do in the body? What body system/organ(s) favors it? What is the expected function for that body system(s)?

* Note: Understanding function helps understand the problems when it is low/high *

- Builds bones + teeth; supports muscle contraction
- Stabilizes neuron excitability; supports heart rhythm
- Required for clotting; regulated by PTH, vitamin D, kidneys

IMPLICATIONS & NURSING MANAGEMENT

What causes it to DECREASE?

- Hypoparathyroidism or thyroidectomy
- Pancreatitis; low vitamin D or magnesium
- High phosphate (including phosphate enemas)
- Loop diuretics/laxatives or chronic GI/renal disease

Signs & Symptoms when it's LOW? Critically LOW?

- + Chvostek / + Trousseau; cramps/tetany
- Laryngospasm/dyspnea; confusion or seizures
- ECG: prolonged QT/ST; severe = ventricular tachycardia
- Slow clotting/bleeding; diarrhea

How do we fix it?

- ECG + airway/neuro; seizure/fall precautions
- IV calcium for severe symptoms as ordered
- Oral calcium + vitamin D when stable
- Monitor Ca/Mg/phosphate; encourage calcium foods

What causes it to INCREASE?

- Hyperparathyroidism
- Malignancy -> calcium released from bone
- Excess calcium antacids / supplements
- Low phosphate

Signs & Symptoms when it's HIGH? Critically HIGH?

- Low HR/RR/BP; weak respirations
- ECG: short QT; widened T wave
- Constipation, N&V; renal stones; bone pain
- Low DTR, weakness, confusion / altered LOC

How do we fix it?

- IV normal saline; monitor urine output + renal function
- Loop diuretic after hydration as ordered
- Monitor ECG/neuro; fall precautions
- Calcitonin as ordered; dialysis if severe

ENCOURAGE

DIETARY SOURCES (HIGHEST CONTENT PER SERVING)

AVOID

Yogurt, milk, sardines w/bones, cheese, cottage cheese, cooked kale, almonds, broccoli, white beans, ice cream.

LOW: ENCOURAGE calcium foods above

HIGH: LIMIT sources above

(arranged from higher content to lower but all rich in calcium.)

HYPOMAGNESEMIA

CRITICAL LOW: not stated; LOW <1.5

MAGNESIUM

RANGE: - 1.5-2.5 mEq/L

HYPERMAGNESEMIA

CRITICAL HIGH not stated; >2.5

What does it do in the body? What body system/organ(s) favors it? What is the expected function for that body system(s)?

* Note: Understanding function helps understand the problems when it is low/high *

- Promotes muscle relaxation; regulates neuromuscular activity

- Supports cardiac rhythm, bones, and immune function

- Helps blood-glucose regulation; closely linked with Ca + K

IMPLICATIONS & NURSING MANAGEMENT

What causes it to DECREASE?

- Alcohol misuse -> poor GI absorption

- NG suction / N&V / diarrhea; diuretics

- Aminoglycoside antibiotics

- Pregnancy-related malnutrition

Signs & Symptoms when it's LOW? Critically LOW?

- Tachycardia/tachypnea; dyspnea

- ECG: long QT, depressed ST, inverted T

- Hyperreflexia/clonus; numbness/tingling; diarrhea

- Confusion/seizures; torsades or VF = danger

How do we fix it?

- ECG; seizure precautions; assess RR + swallowing

- Check Mg/Ca/K and DTR/clonus

- IV magnesium sulfate slowly; recheck labs

- Treat cause; encourage Mg-rich foods

What causes it to INCREASE?

- Renal failure -> decreased Mg excretion

- Excess Mg antacids or laxatives

- Diabetic ketoacidosis

- Addison disease / hyperkalemia

Signs & Symptoms when it's HIGH? Critically HIGH?

- Low BP, RR, and HR

- ECG: widened QRS and prolonged PR

- Low/absent DTR; weakness; hypoactive bowel sounds

- Drowsy/lethargic; coma if severe

How do we fix it?

- Stop Mg products; monitor VS/ECG/RR/LOC/DTR

- IV calcium gluconate as ordered

- Fluids + loop diuretic if renal function allows

- Hemodialysis if severe or renal failure

ENCOURAGE

DIETARY SOURCES (HIGHEST CONTENT PER SERVING)

AVOID

Pumpkin seeds, almonds cooked spinach, black beans, peanut butter, whole-grain cereal, brown rice, oatmeal, avocado

LOW: ENCOURAGE magnesium foods above

HIGH: LIMIT foods + Mg antacids/laxatives

(arranged from higher content to lower but all rich in magnesium.)

HYPOPHOSPHATEMIA

CRITICAL LOW: not stated; LOW <2.5

PHOSPHORUS

RANGE: - 2.5-4.5 mg/dL

HYPERPHOSPHATEMIA

CRITICAL HIGH not stated; >4.5

What does it do in the body? What body system/organ(s) favors it? What is the expected function for that body system(s)?

* Note: Understanding function helps understand the problems when it is low/high *

- Builds bones, teeth, and muscles; stored mainly in bone
- Needed for ATP/energy and cell membranes
- Kidneys + PTH regulate it; phosphate moves opposite calcium

IMPLICATIONS & NURSING MANAGEMENT

What causes it to DECREASE?

- Antacids/malabsorption or low vitamin D
- Hyperparathyroidism / hypercalcemia
- Severe malnutrition or excess alcohol
- Burns, hyperglycemia, or malignancy-related renal loss

Signs & Symptoms when it's LOW? Critically LOW?

- Muscle/respiratory weakness; decreased DTR
- Decreased cardiac output
- Osteomalacia -> fracture/fall risk
- Confusion/seizures; low platelets -> bleeding

How do we fix it?

- Check phosphate/Ca, renal function, ECG + respirations
- Oral phosphate + vitamin D as ordered
- IV phosphate if severe; confirm kidney function
- Monitor calcium; fall/seizure precautions; diet

What causes it to INCREASE?

- Renal insufficiency/failure -> poor excretion
- Fleet / sodium-phosphate enema
- Excess vitamin D or hypoparathyroidism
- Chemotherapy / cell lysis

Signs & Symptoms when it's HIGH? Critically HIGH?

- + Chvostek / + Trousseau; tetany/spasms
- Hyperactive DTR; bone pain
- Laryngospasm = airway danger
- Confusion / mental-status change

How do we fix it?

- Trend phosphate/Ca; ECG/neuro/airway monitoring
- Calcium acetate (PhosLo) WITH FOOD
- Avoid phosphate enemas; limit phosphate foods
- Dialysis if severe or renal failure

ENCOURAGE

DIETARY SOURCES (HIGHEST CONTENT PER SERVING)

AVOID

Pumpkin seeds, salmon, chicken breast, pork chops, Greek yogurt, lentils, milk, peanut butter, black beans, cheese, eggs

LOW: ENCOURAGE phosphate foods above

HIGH: LIMIT sources above

HYPOCHLOREMIA

CRITICAL LOW: not stated; LOW <98

CHLORIDE (Cl-)

HYPERCHLOREMIA

CRITICAL HIGH not stated; >107

RANGE: 98-107 mEq/L*

What does it do in the body? What body system/organ(s) favors it? What is the expected function for that body system(s)?

* Note: Understanding function helps understand the problems when it is low/high *

- Major extracellular anion; follows sodium -> fluid balance

- Balances bicarbonate -> acid-base / pH regulation

- Part of gastric HCl for digestion; kidneys regulate level

IMPLICATIONS & NURSING MANAGEMENT

What causes it to DECREASE?

- Vomiting/NG suction -> loss of gastric HCl

- Loop/thiazide diuretics -> renal chloride loss

- Sweating/burns or excess water/HF/SIADH

- Metabolic alkalosis (bicarbonate rises)

Signs & Symptoms when it's LOW? Critically LOW?

- Usually reflects alkalosis or fluid loss

- Weakness/fatigue; cramps/twitching; tingling

- Confusion/irritability; slow or shallow respirations

- **Dysrhythmia/tetany/seizure if K or Ca also low**

How do we fix it?

- Assess cause + fluid status; VS/neuro/RR/I&O

- Trend Cl/Na/K/HCO₃; ABG if symptomatic

- Volume loss: NS; dilution: fluid restrict as ordered

- Low K: KCl as ordered after renal/urine check

What causes it to INCREASE?

- Large 0.9% saline / excess chloride load

- Severe diarrhea -> bicarbonate loss

- Dehydration/free-water loss: fever, sweating, DI

- Renal dysfunction/RTA or metabolic acidosis

Signs & Symptoms when it's HIGH? Critically HIGH?

- Usually reflects dehydration or metabolic acidosis

- Thirst/dry mucosa; weakness/fatigue; confusion

- Deep, rapid respirations with acidosis

- Low BP/tachycardia/oliguria; edema if fluid overloaded

How do we fix it?

- Trend Cl/Na/K/HCO₃/renal labs; assess lungs/I&O

- Stop/reassess chloride-rich IV; use ordered balanced fluid

- Correct water/volume deficit with ordered fluids

- Treat diarrhea, renal cause, or acidosis

ENCOURAGE

DIETARY SOURCES (HIGHEST CONTENT PER SERVING)

AVOID

Soy sauce, canned soups, deli cold cuts, frozen "tv" dinners, fried foods, pickles, bacon,

LOW: ENCOURAGE salty foods only if prescribed

HIGH: LIMIT table salt + processed foods

(chloride is usually consumed with sodium as table salt.)